Note: No change to any states if NOHIT

/* Bus Operation types */

#define READ 1 /* Bus Read */

#define WRITE 2 /* Bus Write */

#define INVALIDATE 3 /* Bus Invalidate */

#define RWIM 4 /* Bus Read With Intent to Modify */

/* Snoop Result types */

#define NOHIT 0 /* No hit */

#define HIT 1 /* Hit */

#define HITM 2 /* Hit to modified line */

/* L2 to L1 message types */

#define GETLINE 1   /* Request data for modified line in L1 */

#define SENDLINE 2 /* Send requested cache line to L1 */

#define INVALIDATELINE 3 /* Invalidate a line in L1 */

#define EVICTLINE 4

/* Evict a line from L1 */

// this is when L2's replacement policy causes eviction of a line that

// may be present in L1.   It could be done by a combination of GETLINE

// (if the line is potentially modified in L1) and INVALIDATELINE.

Current MESI State	Action	Bus Operation	Snoop Result	Next MESI State	Notes
M (Modified)	Local eviction	FlushWB(Write)	-	I (Invalid)	
	PrRd	-	-	M (Modified)	
	PrWr	-	-	M (Modified)	
	Snooped Upgr/Inv	-	-	-	Not possible
	Snooped Read	FlushWB(Write)	HITM	S (Shared)	
	Snooped ReadX	FlushWB(Write)	HITM	I (Invalid)	
E (Exclusive)	Local eviction	-	-	I (Invalid)	
	PrRd	-	-	E (Exclusive)	
	PrWr	-	-	M (Modified)	
	Snooped Upgr/Inv	-	-	-	Not possible
	Snooped Read	Flush(Write)	HIT	S (Shared)	
	Snooped ReadX	Flush(Write)	HIT	I (Invalid)	
S (Shared)	Local eviction	-	-	I (Invalid)	
	PrRd	-	-	S (Shared)	
	PrWr	Upgr/Inv	-	M (Modified)	

	Snooped Upgr/Inv	-	-	I (Invalid)	
	Snooped Read	-	HIT	S (Shared)	
	Snooped ReadX	-	-	I (Invalid)	
I (Invalid)	Eviction	-	-	-	Not possible
	PrRd	Read	NoHIT	E (Exclusive)	
		Read	HIT/HITM	S (Shared)	
	PrWr	ReadX	-	M (Modified)	
	Snooped Upgr/Inv	-	-	-	
	Snooped Read	-	NoHIT	I (Invalid)	
	Snooped ReadX	-	-	-	

Case analysis

0 read request from L1 data cache

CALL FUNC `process_read_request_data_cach(input logic [ADDRESS_WIDTH-1:0] address)`
Decode the address
readcount ++
Check for lines that have the same tag
    IF Hit, return data to L1,
        MESI doesn't change,
        Update LRU,
        hit counter ++
    ELSE(A MISS)

Miss counter++
Do a BusRd, Snoop Bus
IF HIT, Get data form bus
Find space (can be packed as function)
IF full , call LRU to find a line to evict (message L1 evict),
[In eviction, if M, Bus write, than message L1 evict]
(Cache miss - conflict)
Load in the empty space(fill the cache and MESI),
(Cache miss - unoccoupied)
Change state to S, message L1 sendline

IF HITM Get data from bus,
Find space
Change state to S(Write Back will be handled by the core that has the data), fill line, message L1 sendline

IF NOHIT Read form DRAM
Find space
Change to E, fill line , message L1 sendline

1 write request from L1 data cache

CALL FUNC `process_read_request_instruction_cache(input logic [ADDRESS_WIDTH-1:0] address)`
Decode the Address
Write count++
Check for lines that are valid & same tag
IF write HIT（M,S,E)
Hit count++
IF in S, change to M state
Do bus INVITE
Update LRU
IF in M, remain
Update LRU

IF in E, turn to M
Update LRU

Else write MISS
Miss count++
Bus RWIM
Change state to M , set dirty bit, sendline L1

2 read request from L1 instruction cache

Same as 0

3 snooped read request

Search for the line
Put snoop result (HIT,HITM,NOHIT)
IF HIT , the line in E state -> S, (Requester Read form DRAM)
    S -> S ,(Requester read from DRAM)
IF HITM M ->S, Bus write,
NoHit.

4 snooped write request (Saw an FlushWB,Flush)

“Should” has no effect.

5 snooped read with intent to modify request

Search for the line
Put snoop result (HIT,HITM,NOHIT)
If HIT , E -> I, Invalid command to L1
    S->I, Invalid command to L1 (send text)
If HITM,M ->I , Bus Write(FlushWB), Use **[Getline+inv]**, not [Envict].
NoHit

6 snooped invalidate command

Search for the line
Put snoop result(HIT,HITM,NOHIT)
If HIT E -> I, Invalid command to L1
    S->I, Invalid command to L1 (send text)
If HITM,M ->I , Invalid command to L1
NoHit

8 clear the cache and reset all state

Set all lines to invalid and resetting the statistics

9 print contents and state of each valid cache line

Printout every valid line and calculate the ratio and output